Equilibrium-Based NFT Marketplace Recommendation for NFTs with Breeding

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2025 IEEE International Conference on Data Mining (ICDM) | 979-8-3315-9599-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDM65498.2025.00094

Abstract*—Recently, Non-Fungible Tokens (NFTs) have attracted attention as valuable digital assets. However, NFT marketplaces face complex challenges in simultaneously recommending optimal pricing to sellers and desirable NFTs to buyers. Unlike conventional marketplaces that focus only on balancing demand and supply between sellers and buyers, these tasks are complicated by intricate value interdependencies arising from diverse buyer preferences, budgets, *trait* rarities, and the unprecedented *breeding mechanisms*. This paper formulates the *NFT Project Pricing/Purchasing Recommendation (NP^{3}R)* problem, aiming to achieve a competitive equilibrium that concurrently optimizes seller revenue and buyer utility. We introduce ***BANTER, an iterative algorithm that jointly determines (1) optimal NFT purchases for buyers (via NFT-REC), considering breeding utility and current prices; and (2) optimal pricing for sellers (via PRICE-REC), based on aggregated demand from NFT-REC. To efficiently manage the combinatorial complexity of breeding, we devise *Optimal Parent Pair Selection (OPPS)* and *Heterogeneous Parent Set Selection (HPSS)* schemes. Theoretical analysis guarantees BANTER to converge to a competitive equilibrium. Experiments on five real-world NFT datasets demonstrate its effectiveness in enhancing both seller revenue and average buyer utility. Source code: https://github.com/jimmy-academia/BANTER**

I. INTRODUCTION

Non-fungible tokens (NFTs) have emerged as significant digital assets, representing unique digital art and collectibles.[1] By 2030, the total market value of NFTs is projected to grow up to $80B. NFTs have also seen widespread commercial adoption. *NBA Top Shot*, with total sales exceeding $1B [3], inspired NFT initiatives in other sports leagues [4], [5], [6] and brands, e.g., Nike [7] and Adidas [8]. Moreover, Starbucks' *Odyssey* [9] and Disney's *Pinnacle* [10] further signify interest in NFTs from other market sectors. To accommodate growing demand and participation, *NFT projects*, typically consisting of a limited number of unique NFTs, are distributed through online marketplaces, e.g., OpenSea [11].

This work is supported in part by NSF under grants III-2106758 and POSE-2346158, and in part by Academia Sinica Investigator Project Grant (AS-IV-114-M06), National Science and Technology Council (NSTC113-2221-E-001-016-MY3, NSTC112-2221-E-001-010-MY3, 114-2223-E-002-009), and Taiwan Ministry of Education (114L9009, 114L895504).

[1]Notable examples include Beeple's *"Everydays,"* sold for $69.3M [1], and an *NBA Top Shot* of *"LeBron James Highlight,"* for $208K [2].

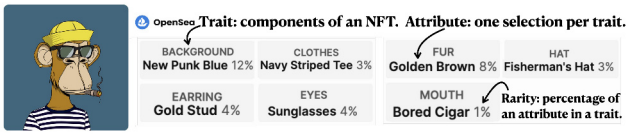


Fig. 1. *BAYC* NFT #4378 with trait rarity displayed on OpenSea [11].

The success of NFT marketplaces hinges on effectively supporting both sellers in pricing their unique creations and buyers in discovering desirable NFTs that align with their preferences and budgets. However, NFTs exhibit unique characteristics that render conventional recommendation and pricing techniques inadequate. First, NFT transactions are transparently recorded on blockchains, in contrast to traditional marketplaces that often rely on opaque or proprietary data [12]. Second, NFTs are inherently non-fungible [13] and derive value from complex trait systems [14], where each token carries attributes, such as color, type, rarity, and lineage, that influence both aesthetic and market value (Fig. (1)). These traits often follow predefined rarity distributions and inheritance logic, shaping user perceptions and strategic preferences. Building on this trait structure, some NFT platforms introduce interactive mechanisms for dynamic value creation, most notably *breeding* [15], [16]. Breeding allows users to combine NFTs to produce new ones, with offspring traits influenced by their digital parents (Fig. (2)). This adds a layer of strategic depth and gamification to ownership [17], creating dynamic interdependencies across the NFT ecosystem, which is an aspect not captured in traditional recommendation settings [18], [19], [20], [21].

In this paper, we formulate the *NP^{3}R (NFT Project Pricing/Purchasing Recommendation)* problem. Our approach is to model NP^{3}R as an economic game, simulating seller and buyer actions within the NFT marketplace with an aim to optimize seller revenue while simultaneously maximizing each buyer's utility of recommended NFTs. In NP^{3}R, the buyers' utility encompasses three key components: (i) the value of owning individual rare NFTs, (ii) the satisfaction derived from forming a personal collection, and (iii) the additional benefits of breeding, according to buyer preferences, budgets, and trait rarity.

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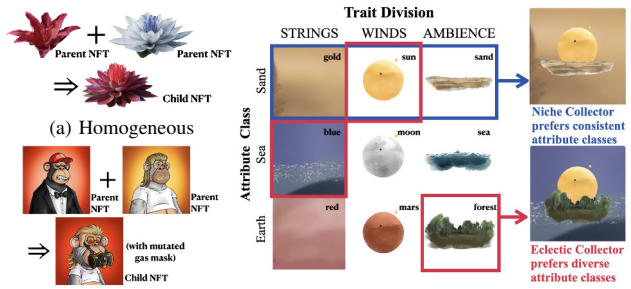


Fig. 2. Examples of NFT project breeding [22], [23], [24] (Section III).

Moreover, NP^{3}R is challenging due to numerous distinct traits and attributes, fixed supply impacting pricing, and complex demand estimation due to breeding. Addressing NP^{3}R also poses significant data mining challenges, including modeling complex NFT attributes, and scaling algorithms for efficient recommendations. We prove that NP^{3}R is computationally hard (PPAD-complete) in Theorem 1.

To solve NP^{3}R problem, we design BANTER, namely, Breeding-aware NFT Equilibrium Recommendation. BANTER leverages a dual-update iterative scheme comprising PRICE-REC and NFT-REC to achieve the *competitive equilibrium*, i.e., the optimal strategy for both the seller and the buyers, respectively (Proposition 1). NFT-REC recommends purchases to optimize buyers' utilities based on pricing derived by PRICE-REC, which then updates the prices of the next iteration based on aggregate demand of all buyers from the current NFT-REC recommendations. Crucially, Theorem 5 establishes BANTER's convergence to the competitive equilibrium, which signifies maximized buyer utilities alongside market-clearing prices that optimize seller revenue.

Central to the complexity of NP^{3}R are the diverse breeding mechanisms supported by popular NFT platforms [11], which enable the creation of new NFTs and significantly impact buyer utility and market dynamics. Understanding these mechanisms is crucial for effective recommendation. This paper considers three primary types of breeding: **Homogeneous Breeding**, which produces offspring by recombining attributes from a pair of parent NFTs (e.g., the new flower in Fig. 2((a)) blending parent attributes); **Child-project Breeding**, which introduces *mutation* and unique child designs (i.e., the "gold mask" on the child ape is mutated in Fig. 2((b)); and **Heterogeneous Breeding**, which forms custom composites from multiple parent NFTs across distinct trait divisions, catering to different collector types (i.e., *niche* and *eclectic*).[2] These mechanisms are formally defined in Section III-B.

To efficiently address the combinatorial challenge in breeding of NFT-REC, we design the *Optimal Parent Pair Selection (OPPS)* for Homogeneous Breeding and Child-project Breeding and *Heterogeneous Parent Set Selection (HPSS)* for Heterogeneous Breeding. Specifically, *OPPS* identifies optimal parent pairs for breeding by evaluating the *amalgamated valuations* (i.e., NFT's personalized value) of buyer preferences and instance rarity, while *HPSS* further curates parent sets by prioritizing those with consistent and diverse attribute classes, preferred by niche and eclectic collectors, respectively. OPPS and HPSS improve BANTER's scalability by significantly reducing computational complexity (Theorem 2), while achieving a bounded approximation ratio (Proposition 2) and maintaining optimal breeding utility for buyers without exhaustive search after convergence.

The contributions in this work are as follows.

- We formulate the NFT Project Pricing/Purchasing Recommendation (NP^{3}R) problem for NFTs with traits and breeding, not explored in traditional recommendations.
- We propose BANTER for NP^{3}R to derive optimal pricing and purchasing recommendations. To address the combinatorial challenge due to breeding, we propose *Optimal Parent Pair Selection (OPPS)* and *Heterogeneous Parent Set Selection (HPSS)* with guaranteed performance.
- Theoretical analyses prove the hardness of NP^{3}R and the convergence of BANTER to a competitive equilibrium.
- Experiments on five real-world NFT projects show that BANTER consistently surpasses all baseline methods in revenue and average utility (with smaller runtime).

II. Related Works

Existing research on pricing explores real-time bidding [25], [26], vehicle dispatch services [27], [28], and seasonal pricing strategies [29].[3] However, these models are not designed for NFTs, which present unique challenges due to their *finite supply*. Similarly, prevalent purchasing recommendation systems primarily emphasize user preferences [31] and item attributes [32], usually overlooking *pricing* and breeding of new NFTs. Such approaches typically employ user-item rating matrices [33], [18], enhance by social network influence [34], [35], [19], and leverage deep learning models [18], [19]. Other studies address recommendation scenarios for item bundles [36], [20] and buyer groups [21], [37], [38] in group recommendations. However, all the above works neglect the *competitive equilibrium* between sellers and buyers, and the interconnectedness of NFTs due to the *trait system* and the *breeding mechanism*, which creates new NFTs not considered in traditional commodity recommendations [18], [19], [20].

Classical recommendation methods fall short because they (1) ignore the critical role of breeding, and (2) treat pricing [28], [29] and purchasing [18], [19] as separate tasks. As such, they cannot support the joint pricing and purchasing recommendations central to NP^{3}R. To our knowledge, no prior work finds competitive equilibrium to jointly optimize seller revenue and buyer utility in NFT markets. Although some studies consider social consensus [37], [38] or item bundling [20], none fully capture the unique characteristics of NFTs—such as trait systems and breeding—and their substantial impact on buyer utility (see Section III-A).

[2]We consider *niche* and *eclectic* collectors who prefer consistency and diversity, respectively; the blue and red boxes in Fig. 2((c)) illustrate corresponding selections of uniform and varied attribute classes for breeding.

[3]This section reviews related works in pricing and purchasing recommendations. For works on blockchain and other types of recommendations, see [30].

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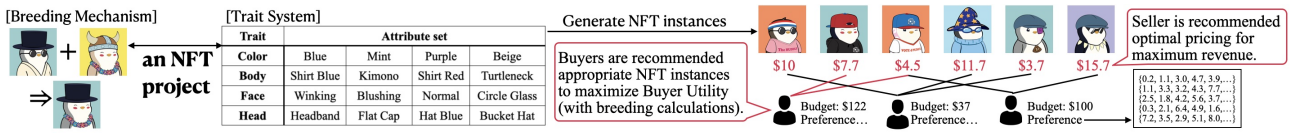


Fig. 3. Illustration of the NFT Project Pricing/Purchasing Recommendation (NP^{3}R) problem.

TABLE I
SUMMARY OF NOTATIONS (FULL TABLE IN [30]).

Notations	Descriptions
$\mathcal{N}$	The set of N NFT buyers, denoted as $b_1, \ldots, b_N$.
$\mathbf{a}^i$	Trait affinity tensor of buyer b_i, consisting of T affinity vectors, $\mathbf{a}^i = (\mathbf{a}^i_1, \ldots, \mathbf{a}^i_T)$.
$\mathcal{M}$	The set of M NFT instances $\eta_1, \ldots, \eta_M$.
$\boldsymbol{\alpha}^j$	Attribute tensor of NFT instance η_j, consisting of T one-hot vectors, $\boldsymbol{\alpha}^j = (\boldsymbol{\alpha}^j_1, \ldots, \boldsymbol{\alpha}^j_T)$.
B^i, U^i, R^i	Budget, utility, and remaining budget of b_i.
$\mathcal{A}_t$	The attribute set for each trait $t \in [1, \ldots, T]$.
$\mathbf{Q}$	The supply vector. $\mathbf{Q}_j$ records the number of copies of NFT assets per NFT instance η_j.
$\mathbf{p}$	The recommended pricing ($\mathbf{p}_j$ is price of η_j).
$\mathbf{x}^i$	The recommended purchases for buyer b_i. $\mathbf{x}^i_j$ indicates the amount of η_j recommended to b_i.
$\mathcal{P}, \mathcal{L}, \mathcal{C}$	A parent NFT set, pruned list of parent NFTs, and the candidate set of parent NFT sets.
$V(\eta_j)$	The objective valuation of η_j.
$\tilde{V}^i(\eta_j)$	The amalgamated valuation of η_j for buyer b_i.
$q(\boldsymbol{\alpha}^j_t)$	The number of NFTs possessing attribute $\boldsymbol{\alpha}^j_t$.
k	The breeding count restriction.
$\mathcal{K}$	The subset of that yields the top-k breeding results.
r	The random mutation rate.
r^i_{breed}	The child breeding rate for buyer i.
f_{pop}	The population factor due to market saturation.
$\mathbf{z}$	The excess demand.

III. THE NP^{3}R PROBLEM

NP^{3}R includes N buyers and M NFT instances in an NFT project with a *trait system* and a *breeding mechanism*. Table I summarizes the notations.

Definition 1 (Trait System)**.** *A trait system comprises T traits, where each NFT instance η_j is defined by an attribute tensor $\boldsymbol{\alpha}^j = (\boldsymbol{\alpha}^j_1, \ldots, \boldsymbol{\alpha}^j_T)$, specifying one attribute per trait. Each instance η_j has a supply of $\mathbf{Q}_j$ interchangeable copies, each referred to as an* NFT asset.

Definition 2 (Breeding Mechanism)**.** *The NFT breeding mechanism enables the creation of a new child NFT, $\eta_c = \text{BREED}(\mathcal{P})$, from parent NFTs $\mathcal{P}$, under a breeding count limit of k child NFTs per buyer, which aims to prevent market oversupply in order to retain the NFT value.*

NP^{3}R aims to find the optimal pricing recommendation to the seller and the optimal purchasing recommendation to the buyers simultaneously. As shown in Fig. 3, NP^{3}R comprise two subproblems: 1) *Buyer Purchasing Recommendation (BPR)*, which recommends NFT purchases to maximize the utilities of buyers according to their budgets and preferences, NFT pricing, and the breeding mechanism; 2) *Seller Pricing Recommendation (SPR)*, which optimizes pricing for maximum seller revenue. We use $\mathbf{x}^i$ to formally represent buyer b_i's purchasing recommendations under the pricing vector $\mathbf{p}$, where $\mathbf{x}^i_j$ (jth element of $\mathbf{x}^i$) is the amount of NFT η_j recommended to b_i and $\mathbf{p}_j$ (jth element of $\mathbf{p}$) is the price of η_j.

Definition 3 (The BPR problem)**.** *Given a pricing $\mathbf{p}$ and $U^i(\mathbf{x}^i)$, the utility of buyer b_i, the BPR problem aims to solve*

$$\hat{\mathbf{x}}^i = \arg\max_{\mathbf{x}^i} U^i(\mathbf{x}^i),\ s.t.\ \mathbf{x}^i \cdot \mathbf{p} \le B^i, \tag{1}$$

where B^i is the budget of buyer b_i.

Definition 4 (The SPR problem)**.** *Given the NFT set $\mathcal{M}$ with supply $\mathbf{Q}$ and the buyer set $\mathcal{N}$, SPR seeks to maximize*

$$\hat{\mathbf{p}} = \arg\max_{\mathbf{p}} \sum_{\eta_j \in \mathcal{M}} \mathbf{p}_j \cdot \min\left(\mathbf{Q}_j, \sum_{b_i \in \mathcal{N}} \mathbf{x}^i_j\right) \tag{2}$$

according to the solution $\mathbf{x}^i$ of BPR, where $\min(\cdot)$ restricts total purchase from exceeding over the supply.

The hardness of NP^{3}R is presented as follows.

Theorem 1. *(Proved in [30]) The NP^{3}R problem for M instances and N buyers with budgets is PPAD-complete.*

As shown, the joint optimization of BPR and SPR presents significant computational challenges. A buyer's decision space spans M dimensions, corresponding to potential budget allocations across M NFT instances, and the sheer scale and combinatorial complexity of the joint action space for these N buyers are primary drivers of the problem's PPAD-completeness. Therefore, our strategy is to find the competitive equilibrium between buyers and sellers [39] by individually deriving the optimal solutions of BPR and SPR to iteratively update the pricing and the purchasing recommendations.

A. Buyer's Utility

Following the literature on buyer behaviors [40] and NFT valuation [41], we introduce key components of buyer utility in the NFT market. Beyond the value of money (represented by the *remaining budget* R^i) [42], a buyer b_i's utility comprises 1) *Instance-wise NFT utility* ($U^i_{Instance}$), representing the objective value of individual NFTs, primarily determined by their rarity within the market [14]; 2) *Collection-based NFT utility* ($U^i_{Collection}$), capturing the subjective appreciation derived from owning a set of NFTs, considering the interrelationships and collective attributes of the NFTs in a buyer's portfolio [40]; and 3) *NFT breeding utility* ($U^i_{Breeding}$), accounting for the potential value of child NFTs generated via the breeding mechanism [15], [16]. Existing recommendation systems, however, largely overlook such multifaceted NFT utility, particularly the value derived from breeding and collections [18], [43].

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Definition 5 (Buyer's Utility)**.** *The buyer's utility U^i for b_i is*

$$U^i(\mathbf{x}^i) \equiv U^i_{Instance}(\mathbf{x}^i) + U^i_{Collection}(\mathbf{x}^i) + U^i_{Breeding}(\mathbf{x}^i) + R^i, \quad (3)$$

where R^i is the remaining budget of buyer b_i.

Instance-wise NFT Utility ($U^i_{Instance}$). It captures the objective value of individual NFTs, primarily driven by attribute rarity [41], a factor buyers actively consider [14]. Formally,

$$U^i_{Instance}(\mathbf{x}^i) \equiv \sum_{\eta_j \in \mathcal{M}} \mathbf{x}^i_j V(\eta_j),\ V(\eta_j) = \sum_{t=1}^{T} \log\left(\frac{|\mathbf{Q}|}{q(\boldsymbol{\alpha}^j_t)}\right), \quad (4)$$

where $V(\eta_j)$ is the objective valuation of NFT η_j. Following [14], $q(\boldsymbol{\alpha}^j_t)$ is the number of NFTs with attribute $\boldsymbol{\alpha}^j_t$, and $|\mathbf{Q}|$ is the total count, modeling the inverse relationship between attribute frequency and value, crucial given NFTs' non-fungible nature and inherent supply limit $\mathbf{Q}$ for rarity.

Collection-based NFT Utility ($U^i_{Collection}$). Collectors derive utility from curating portfolios based on personal attribute preferences and inter-NFT synergy [44], supported by marketplace features [11]. Formally,

$$U^i_{Collection}(\mathbf{x}^i) \equiv \sum_{t=1}^{T} \mathbf{a}^i_t \cdot \log\left(\sum_{\eta_j \in \mathcal{M}} \mathbf{x}^i_j \boldsymbol{\alpha}^j_t + \mathbf{1}\right), \quad (5)$$

where $\mathbf{a}^i_t$ is buyer b_i's trait affinity tensor. $U^i_{Collection}$ thus reflects the utility from the buyer's aggregated NFT attributes, with diminishing returns (via the logarithm) on attribute sums, weighted by individual preferences.

NFT Breeding Utility ($U^i_{Breeding}$) is a unique feature not considered in existing recommendations [18], [19]. Drawing from fractional NFT concepts [45], [46], we define the child breeding rate $r^i_{breed}(\mathcal{P})$ which represents the amount of child NFTs generated by combining the parent set $\mathcal{P}$ with fractional parent NFTs [47], [48] as

$$r^i_{breed}(\mathcal{P}) \equiv \frac{1}{|\mathcal{P}|} \sum_{\eta_j \in \mathcal{P}} \min(1, \mathbf{x}^i_j), \quad (6)$$

where each parent NFT's contribution is proportional to its ownership share $\mathbf{x}^i_j$. By incorporating $r^i_{breed}(\mathcal{P})$, buyers are incentivized to increase their purchasing on NFTs for breeding.

Definition 6. *The NFT breeding utility is defined as*

$$U^i_{Breeding} \equiv \max_{\mathcal{K}^i \subset \mathcal{S}} \sum_{\mathcal{P} \in \mathcal{K}^i} r^i_{breed}(\mathcal{P}) \mathbb{V}^{i,\text{TYPE}}[\mathcal{P}], \quad (7)$$

where $\mathcal{K}^i$ is a subset of all parent combinations $\mathcal{S}$ whose total child breeding rate $\sum_{\mathcal{P} \in \mathcal{K}^i} r^i_{breed}(\mathcal{P})$ is within the breeding count limit k (Definition 2). $\mathbb{V}^{i,\text{TYPE}}[\mathcal{P}]$ is buyer b_i's expected value of a child NFT from parent set $\mathcal{P}$ under different breeding types ("homo," "child," "niche," "eclectic"), defined later in Definitions 7 to 9.

Example. As shown in Fig. 2, different breeding mechanisms yield distinct results. Homogeneous Breeding (Fig. 2((a))) blends parental traits for refined value, while Child-project Breeding (Fig. 2((b))) introduces mutations (e.g., unique glasses) for novelty and rarity. Heterogeneous Breeding (Fig. 2((c))) further create composites from uniform attribute classes (e.g., 'Sand') for thematic consistency appealing to *niche* collectors, or from varied classes for *eclectic* collectors.

B. The Expected Value Functions for Breedings

To account for market saturation from duplicated NFT attributes in Homogeneous Breeding, we adopt the *population factor* $f_{pop}(\eta_c) = \sum_{t=1}^{T} \exp\left(-\frac{q(\boldsymbol{\alpha}^c_t)}{|\mathbf{Q}|}\right)$, following [14]. Here, $f_{pop}(\eta_c)$ considers each attribute $\boldsymbol{\alpha}^c_t$ of child NFT η_c, where $q(\boldsymbol{\alpha}^c_t)$ is the number of NFTs with $\boldsymbol{\alpha}^c_t$, and $|\mathbf{Q}|$ is the total NFT count. Furthermore, we introduce the buyer's amalgamated valuation [14], [41], $\tilde{V}^i(\eta) = \left(\sum_t \mathbf{a}^i_t \cdot \boldsymbol{\alpha}_t\right) V(\eta)$, tailoring the objective valuation of $V(\eta)$ by buyer preferences.

Definition 7. *$\mathbb{V}^{i,homo}$ for Homogeneous Breeding is*

$$\mathbb{V}^{i,homo}[\mathcal{P}] \equiv \mathbb{E}\left[f_{pop}(\eta_c)\tilde{V}^i(\eta_c)\right], \eta_c = \text{BREED}(\mathcal{P}), \quad (8)$$

where $\tilde{V}^i$ is the amalgamated valuation, and each child attribute $\boldsymbol{\alpha}^c_t$ is inherited from parent p_1 or p_2 (from $\mathcal{P}$) with equal probability.

Definition 8. *$\mathbb{V}^{i,child}$ for Child-project Breeding is*

$$\mathbb{V}^{i,child}[\mathcal{P}] \equiv \mathbb{E}\left[\tilde{V}^i(\eta_c)\right], \eta_c = \text{CHILD-BREED}(\mathcal{P}, r), \quad (9)$$

where CHILD-BREED *creates η_c such that each attribute $\boldsymbol{\alpha}^c_t$ is assigned $\boldsymbol{\alpha}^{p_1}_t$ or $\boldsymbol{\alpha}^{p_2}_t$ (from $\mathcal{P}$) with probability $(1-r)/2$ each;[4] otherwise, $\boldsymbol{\alpha}^c_t$ is randomly assigned an attribute from $\mathcal{A}_t$ with mutation probability r for each trait t.[5]*

Finally, Heterogeneous Breeding categorizes NFTs using *trait divisions*, allowing buyers to select one parent NFT from each division, and *attribute classes*, which guide design style choices to cater to distinct (niche or eclectic) preferences.

Definition 9. *In Heterogeneous Breeding, the parent NFT set $\mathcal{P}$ consists of NFT instances from each of the D trait divisions, with $|\mathcal{P}| = D$. For niche collectors, the expected value is*

$$\mathbb{V}^{i,niche}[\mathcal{P}] = \mathbb{M}(\mathcal{P}) \sum_{\eta \in \mathcal{P}} \tilde{V}^i(\eta), \quad (10)$$

where $\mathbb{M}(\mathcal{P}) \equiv \max_c |\{\eta \in \mathcal{P} \mid class(\eta) = c\}|$ returns the count of NFTs in the most frequent attribute class c within $\mathcal{P}$, and $\tilde{V}$ is the amalgamated valuation. For eclectic collectors,

$$\mathbb{V}^{i,eclectic}[\mathcal{P}] = \mathbb{U}(\mathcal{P}) \sum_{\eta \in \mathcal{P}} \tilde{V}^i(\eta), \quad (11)$$

where $\mathbb{U}(\mathcal{P}) \equiv |\{c \mid \exists \eta \in \mathcal{P}, class(\eta) = c\}|$ counts the distinct attribute classes among $\eta \in \mathcal{P}$.

Here, $\mathbb{M}$ is a *majority count* favoring the majority attribute class in $\mathcal{P}$, suiting *niche* collectors who prefer consistent breeding designs. Conversely, the *diversity count* $\mathbb{U}$ favors attribute class variety in $\mathcal{P}$, supporting *eclectic* collectors' preference for diverse designs. This dual expectation definition under Heterogeneous Breeding underscores the need for customized recommendations for specific buyer preferences.

[4]Following *CryptoKitties* [49], r is predefined by the sellers.

[5]Child-project Breeding set all $f_{pop} = 1$ as it addresses market saturation by creating unique child NFT designs and through mutation (with r) that can introduce rare attributes not found in parent NFTs.

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Algorithm 1 The BANTER method.

Input: Budgets $\{B^i\}$, affinity tensors $\{\mathbf{a}^i\}$, attribute tensors $\{\boldsymbol{\alpha}^j\}$ supply vector $\mathbf{Q}$, step-size ϵ, parameters K_{init}, K, K_d. Candidate set length K_c. TYPE(i) is the breeding and collector type of b_i.

$\mathbf{p} \leftarrow$ INIT$(K_{init}, \{\mathbf{a}^i\}, \{\boldsymbol{\alpha}^j\}, \{B^i\}, \mathbf{Q})$
for K iterations **do**
  for $i = 1$ **to** N **do**
    $\mathbf{x}^i \leftarrow$NFT-REC $(K_d, \mathbf{a}^i, \{\boldsymbol{\alpha}^j\}, B^i, \mathbf{p}, K_c,$ TYPE$(i))$
  $\epsilon, \mathbf{p} \leftarrow$PRICE-REC $(\epsilon, \mathbf{p}, \{\mathbf{x}^i\}, \mathbf{Q})$
procedure INIT$(K_{init}, \{\mathbf{a}^i\}, \{\boldsymbol{\alpha}^j\}, \{B^i\}, \mathbf{Q})$
  random initialize $\mathbf{p}$
  for K_{init} iterations **do**
    $\forall i, j : \mathbf{s}^i_j \leftarrow \sum_t \mathbf{a}^i_t \boldsymbol{\alpha}^j_t / \mathbf{p}_j$
    $\forall j : \mathbf{p}_j \leftarrow \left(\sum_i B^i \cdot \frac{\mathbf{s}^i_j}{\sum_j \mathbf{s}^i_j} \right) / \mathbf{Q}_j$
  return $\mathbf{p}$

IV. BANTER

We present *BANTER*, Breeding-aware NFT Equilibrium Recommendation, an iterative algorithm featuring dual recommendation components: pricing (PRICE-REC) and NFT purchasing with breeding (NFT-REC), to address NP^{3}R. BANTER models NP^{3}R as an economic game, simulating seller and buyer actions within the NFT marketplace. The goal is a competitive equilibrium that achieves optimal buyer utility and seller revenue [50]. Detailed examples are provided in [30].

Definition 10. *A competitive equilibrium is a strategy profile* $(\mathbf{p}^*, \mathbf{x}^{i*})$ *where i)* $\mathbf{p}^*$ *is the optimal pricing that solves the SPR problem, and ii)* $\mathbf{x}^{i*}$ *is the optimal purchasing for each buyer* b_i *that solves the BPR problem, given* $\mathbf{p}^*$.

Since pricing and purchasing are interdependent, iterative updates are required to reach equilibrium. Crucially, we show that this competitive equilibrium is obtained at the *market-clearing condition* [51], [52], where supply matches demand, ensuring full allocation without surplus or shortage.

Proposition 1. *A market-clearing condition is the joint recommendations* $\{\tilde{\mathbf{p}}, \tilde{\mathbf{x}}\}$ *where*

$$\forall b_i \in \mathcal{N} : \tilde{\mathbf{x}}^i = \underset{\mathbf{x}^i \cdot \tilde{\mathbf{p}} \le B^i}{\arg\max} U^i(\mathbf{x}^i) \ , \ \textit{s.t.} \ \forall j : \sum_{i=1}^{N} \tilde{\mathbf{x}}^i_j = \mathbf{Q}_j, \quad (12)$$

establishes a competitive equilibrium (proof in [30]).

The connection between the market-clearing condition and competitive equilibrium provides the theoretical foundation for BANTER. By designing our algorithm to attain NFT market clearing, we solve NP^{3}R with two main components (Algorithm 1). PRICE-REC refines pricing (to solve SPR) by i) adjusting prices of high-demand NFT instances after buyer recommendations from NFT-REC, and ii) designing a *demand-aware* step-size schedule to accelerate convergence. NFT-REC then finds optimal buyer purchases (to solve BPR) by i) recommending NFTs based on the updated pricing from PRICE-REC, and ii) selecting appropriate breeding combinations. The adaptive interaction between PRICE-REC and NFT-REC ensures that BANTER converges to market clearing, as proven in Theorem 5. Moreover, to speed up convergence, we design INIT for BANTER to establish initial preference-aware prices, thereby providing a more refined starting point for BANTER's main iterative process, by iteratively improving them over K_{init} iterations, based on buyer preferences ($\mathbf{a}^i$) and NFT attributes ($\boldsymbol{\alpha}^j$). Afterward, BANTER alternates between improving pricing (via PRICE-REC) and purchasing recommendations (via NFT-REC) to progressively converge towards equilibrium. To efficiently manage the complex breeding utility calculations within NFT-REC, BANTER also employs specialized pruning schemes, i.e, *Optimal Parent Pair Selection (OPPS)* for Homogeneous Breeding and Child-project Breeding, and *Heterogeneous Parent Set Selection (HPSS)* for Heterogeneous Breeding. These schemes identify high-potential parent NFTs by considering buyer preferences, attribute rarity, and buyer styles (*niche* or *eclectic*), thereby reducing the cost of exploring all possible breeding options.

Algorithm 2 NFT-REC and PRICE-REC for a buyer b_i.

procedure NFT-REC$(K_d, \mathbf{a}^i, \{\boldsymbol{\alpha}^j\}, B^i, \mathbf{p}, K_c,$ TYPE$)$
  random init $|\mathbf{s}^i| = 1$, $\forall j : \mathbf{x}^i_j \leftarrow (\mathbf{s}^i_j B^i)/\mathbf{p}_j$
  for K_d iterations **do**
    $U^i_{Instance} \leftarrow \sum_j \mathbf{x}^i_j \cdot V(\boldsymbol{\alpha}^j)$
    $U^i_{Collection} \leftarrow \sum_t \mathbf{a}^i_t \cdot \log\left(\sum_j \mathbf{x}^i_j \boldsymbol{\alpha}^j_t + 1\right)$
    $U^i_{Breeding} \leftarrow$ BREEDING-UTILITY$(\mathbf{x}^i, K_c,$ TYPE$)$
    $U^i \leftarrow U^i_{Instance} + U^i_{Collection} + U^i_{Breeding} + B^i \mathbf{s}^i_{-1}$
    $\mathbf{s}^i \leftarrow \left(\mathbf{s}^i + \epsilon_s \frac{\partial U^i}{\partial \mathbf{s}}\right)$; $\mathbf{s}^i \leftarrow \frac{\mathbf{s}^i}{|\mathbf{s}^i|}$; $\forall j : \mathbf{x}^i_j \leftarrow (\mathbf{s}^i_j B^i)/\mathbf{p}_j$
  return $\mathbf{x}^i$
procedure PRICE-REC$(\epsilon, \mathbf{p}, \{\mathbf{x}^i\}, \mathbf{Q})$
  $\mathbf{z} \leftarrow \sum_i \mathbf{x}^i - \mathbf{Q}$; $\epsilon \leftarrow \epsilon \exp\left(\gamma \frac{\|\mathbf{z}\|_2}{\|\mathbf{Q}\|_2}\right)$; $\mathbf{p} \leftarrow \mathbf{p}\left(1 + \epsilon \frac{\mathbf{z}}{|\mathbf{z}|}\right)$
  return $\epsilon, \mathbf{p}$

A. The NFT-REC Procedure

Given a pricing $\mathbf{p}$ obtained in the previous iteration of PRICE-REC, NFT-REC solves BPR by finding the purchase recommendation $\mathbf{x}^i$ for each buyer b_i that maximizes their individual utility U^i (Equation (3)). As detailed in Algorithm 2, NFT-REC first introduces an expenditure proportion vector $\mathbf{s}^i \in [0,1]^{M+1}$ for each buyer b_i, normalized with its element sum as one (i.e., $\sum_k \mathbf{s}^i_k = 1$). Each element $\mathbf{s}^i_j$ (for $j \neq -1$) represents the fraction of budget B^i allocated to purchase NFT instance η_j, while a special element $\mathbf{s}^i_{-1}$ represents the fraction of the budget to be retained ($R^i \equiv \mathbf{s}^i_{-1} B^i$). Based on $\mathbf{s}^i$,$\mathbf{x}^i_j \equiv B^i \mathbf{s}^i_j / \mathbf{p}_j$ for NFT η_j.

NFT-REC iteratively refines $\mathbf{s}^i$ over many iterations. In each iteration, it calculates the instance-wise ($U^i_{Instance}$) and collection-based ($U^i_{Collection}$) utility components using the current $\mathbf{x}^i$. It then derives the breeding utility ($U^i_{Breeding}$) according to the BREEDING-UTILITY procedure (Section IV-C), which carefully examines diverse breeding configurations. To improve the total utility U^i, $\mathbf{s}^i$ is updated using gradient ascent with respect to U^i and then re-normalized (L1-norm) to maintain in budget. Thus, NFT-REC aims to maximize each buyer's utility U^i while inherently adhering to their budget B^i. In contrast to NFT-REC, traditional commodity recommendation

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Algorithm 3 The BREEDING-UTILITY calculation.

procedure BREEDING-UTILITY($\mathbf{x}^i, K_c$, TYPE)
  if TYPE **is** homo **or** child **then** $\mathcal{C} \leftarrow$ OPPS (i, K_c)
  if TYPE **is** niche **or** eclectic **then** $\mathcal{C} \leftarrow$ HPSS (i, K_c)
  $c_{breed} \leftarrow 0$; $U^i_{Breeding} \leftarrow 0$
  for $\mathcal{P}$ **in** $\mathcal{C}$ **do**
    $r_{breed} \leftarrow \frac{1}{|\mathcal{P}|} \sum_{\eta_j \in \mathcal{P}} \min(\mathbf{x}^i_j, 1)$
    $c_{breed} \leftarrow c_{breed} + r_{breed}$
    $U^i_{Breeding} \leftarrow U^i_{Breeding} + r^i_{breed} \mathbb{V}^{i,\text{TYPE}}[\mathcal{P}]$
    if $c_{breed} \geq k$ **then break**
  return $U^i_{Breeding}$

Algorithm 4 OPPS (Optimal Parent Pair Selection)

procedure OPPS(i, K_c)
  $\mathcal{L} \leftarrow$ top K_c instances from $\mathcal{M}$ **sorted by** $f_{pop}(\eta_j)\tilde{V}^i(\eta_j)$
  $\mathcal{C} \leftarrow \{(\eta_p, \eta_q) \mid \eta_p, \eta_q \in \mathcal{L}, \eta_p \neq \eta_q\}$
  sort $\mathcal{C}$ **by** $\frac{1}{2}\sum_{\eta \in \mathcal{P}} f_{pop}(\eta_p)\tilde{V}^i(\eta)$
  return $\mathcal{C}$

Algorithm 5 HPSS (Heterogeneous Parent Set Selection)

procedure HPSS(i, K_c)
  $\mathcal{C} \leftarrow \{\}$; $\mathcal{L} \leftarrow$ top K_c instances from $\mathcal{M}$ **sorted by** $\tilde{V}^i(\eta_j)$
  for η_j **in** $\mathcal{L}$ **do**
    $\mathcal{P} \leftarrow \{\eta_j\}$
    for k **from** j **to** K_c **do**
      if $trait_division(\eta_k)$ **not in** $\mathcal{P}$ **then** $\mathcal{P}$ **append** η_k
      if $|\mathcal{P}| = D$ **then** $\mathcal{C}$ **append** $\mathcal{P}$; **break**
  $\mathbb{F} \equiv \mathbb{M}$ **if** b_i **is** niche collector **else** $\mathbb{F} \equiv \mathbb{U}$
  sort $\mathcal{C}$ **by** $\mathbb{F}(\mathcal{P})\sum_{\eta \in \mathcal{P}} \tilde{V}^i(\mathcal{P})$
  return $\mathcal{C}$

systems do not incorporate dynamic pricing into purchase suggestions [18], [43], [32] and, as they are designed for non-generative items, inherently omit considerations of NFT-specific breeding [18], [43], [32], [33], [31].

B. The PRICE-REC Procedure

In each iteration t, PRICE-REC improves the pricing $\mathbf{p}^t$ by

$$\mathbf{p}^{t+1} = \mathbf{p}^t \left(1 + \epsilon^t \frac{\mathbf{z}}{\|\mathbf{z}\|_2}\right), \ \mathbf{z} \equiv \sum_i \mathbf{x}^i(\mathbf{p}^t) - \mathbf{Q}, \tag{13}$$

where $\mathbf{z}$ is the *excess demand*, i.e., the difference between the sum of $\mathbf{x}$ (recommendation from NFT-REC) and $\mathbf{Q}$. In particular, prices are raised for NFT η_j when $\mathbf{z}_j > 0$ and vice versa. To expedite convergence, we design an adaptive *demand-aware* scheduling,

$$\epsilon^t = \epsilon^{t-1} \exp\left(\frac{\|\mathbf{z}\|_2}{\|\mathbf{Q}\|_2}\right), \tag{14}$$

where ϵ^{t-1} is the step size in the previous iteration. Intuitively, we increase the step size when the overall market imbalance (i.e., the normalized magnitude of total excess demand, $\frac{\|\mathbf{z}\|_2}{\|\mathbf{Q}\|_2}$) becomes substantial to support a more decisive price adjustment to rapidly correct significant demand-supply disparities and thereby expedite convergence towards equilibrium.

C. The BREEDING-UTILITY Improvement

Optimizing $U^i_{Breeding}$ by brute-force examination of all parent combinations $\mathcal{P}$ in BREEDING-UTILITY (Algorithm 3) is computationally prohibitive. To enhance scalability, we design *Optimal Parent Pair Selection (OPPS)* (Algorithm 4) for Homogeneous Breeding and Child-project Breeding, and *Heterogeneous Parent Set Selection (HPSS)* (Algorithm 5) for Heterogeneous Breeding. OPPS and HPSS identify a candidate set $\mathcal{C}$ of promising parent NFT sets by selectively evaluating preferences across various breeding and buyer types to reduce computational overhead. Illustrations of OPPS and HPSS is presented in [30].

Specifically, since $U^i_{Breeding}$ (Definition 6) prioritizes high-value parent combinations, pruning less valuable pairs from the search space is important. Parent candidates are first sorted for each buyer to evaluate the influence of buyer-specific preferences ($\tilde{V}^i$) on breeding utility. OPPS ranks NFTs by $f_{pop}(\eta_j)\tilde{V}^i(\eta_j)$ for Homogeneous Breeding and by $\tilde{V}^i(\eta_j)$ for Child-project Breeding, selecting the top K_c for a candidate list $\mathcal{L}$ to reduce the search space. According to f_{pop}, it yields a competitor-aware list prioritizing rarer outcomes. Candidate set $\mathcal{C}$ then includes all parent pairs $\mathcal{P}$ formed from $\mathcal{L}$, ranked by their combined $\tilde{V}^i(\eta_p) + \tilde{V}^i(\eta_q)$.

For Heterogeneous Breeding, HPSS selects the top-K_c list $\mathcal{L}$ from $\mathcal{M}$ based on $\tilde{V}^i$. Parent sets $\mathcal{P} \in \mathcal{C}$ are formed by selecting one parent from $\mathcal{L}$ and completing the set with others from $\mathcal{L}$ to carefully include all trait divisions. These sets $\mathcal{P}$ are then ranked by attribute class consistency ($\mathbb{M}(\mathcal{P})$) or diversity ($\mathbb{U}(\mathcal{P})$) to suit niche or eclectic collectors, respectively (Definition 9), identifying high-value, tailored parent sets to effectively accelerate BANTER's convergence (Proposition 2).

From candidate set $\mathcal{C}$, an optimal subset $\mathcal{K}^i \subseteq \mathcal{C}$ is chosen for buyer b_i. $\mathcal{K}^i$ maximizes $\sum_{\mathcal{P} \in \mathcal{K}^i} r^i_{breed}(\mathcal{P})\mathbb{V}^{i,\text{TYPE}}[\mathcal{P}]$ with the total breeding rate $\sum_{\mathcal{P} \in \mathcal{K}^i} r^i_{breed}(\mathcal{P}) \leq k$ (the breeding count limit). The selection involves iteratively adding parent pairs from the ranked candidate set $\mathcal{C}$ to $\mathcal{K}^i$, ensuring the total breeding rate remains within the limit k. For efficient calculation of $f_{pop}(\eta_c)$ used in $\mathbb{V}^{i,homo}$ for Homogeneous Breeding, we exploit a hash map to track processed parent sets per buyer. Global attribute counts $q(\boldsymbol{\alpha}^c_t)$ are updated by aggregating anticipated child attributes from all buyers' current breeding decisions (weighted by their $r^i_{breed}(\mathcal{P})$).

The overall complexity reduction of BANTER by OPPS and HPSS is presented as follows.

Theorem 2. *By applying OPPS under Homogeneous Breeding and Child-project Breeding, BANTER's time complexity is reduced from $O(KNM^2)$ to $O(KN|\mathcal{C}|)$, where K is the iteration count, $|\mathcal{C}|$ is the constant candidate set length, and N, M, and D is the number of buyers, NFT instances, and trait divisions, respectively. Similarly, applying HPSS under Heterogeneous Breeding lowers the time complexity from $O(KNM^D)$ to $O(KN|\mathcal{C}|)$ (proved in [30]).*

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V. Theoretical Analysis

Here, we prove that BANTER converges to the competitive equilibrium for NP^{3}R, where the equilibrium is connected to the market clearing condition based on Proposition 1.

BANTER's convergence is proven by a) leveraging an aggregate utility to ensure the resulting allocation is optimal for every individual buyer (Theorem 3), b) establishing an excess demand potential function whose gradient provides a clear, computable signal for adjusting prices towards market-clearing (Theorem 4), and c) using this gradient scheme to guarantee the iterative process reaches a stable competitive equilibrium (Theorem 5). This framework is designed to be comprehensive, as the buyer utility that underpins this entire analysis explicitly incorporates the nuanced values derived from NFT breeding. Besides, we design the demand-aware step-size scheduling (Equation (14)). Intuitively, this allows PRICE-REC to make larger price adjustments when the market is far from equilibrium and smaller, more careful adjustments as it gets closer, thus accelerating progress while preventing overshooting. The effectiveness of this acceleration is empirically verified in our ablation study in Section VI-C.

Aggregated Utility. We first prove that an optimal solution for an aggregated utility U aligns with the optimal solution for $U^i(\mathbf{x}^i)$ of each buyer b_i. U is defined as follows.

$$U \equiv \prod_i U^i(\mathbf{x}^i)^{\beta^i} \text{ , and } \beta^i \equiv B^i / \sum_i B^i, \tag{15}$$

where the normalized budget exponent β^i becomes a scalar weight in $\log(U)$, and finds the solution

$$\bar{\mathbf{x}}^1, \ldots, \bar{\mathbf{x}}^N \in \underset{\mathbf{x}^1,\ldots,\mathbf{x}^N}{\arg\max} U, \ s.t. \sum_i \mathbf{x}^i \leq \mathbf{Q}. \tag{16}$$

Theorem 3. *Assuming U^i (and thus U) are concave, homogeneous, continuous, and non-decreasing utility functions, the aggregated solution $\bar{\mathbf{x}}^i$ to Equation (16) is an optimal solution to Equation (12). (Please see the proof in [30].)*

Excess Demand Potential. To assess the deviation from equilibrium before convergence, we introduce the excess demand potential ϕ using the Lagrangian of $\log U = \sum_i \beta_i \log U^i(\mathbf{x}^i)$, where $\log$ is applied for a simplified representation [39]. The optimization problem of $\log U$ under the supply constraint (Equation (12)) yields the Lagrangian formulation below.

$$\mathscr{L}(\{\mathbf{x}^i\}, \mathbf{p}) \equiv \sum_i \beta^i \log U^i(\mathbf{x}^i) - \sum_j \mathbf{p}_j (\sum_i \mathbf{x}^i_j - \mathbf{Q}_j), \tag{17}$$

where $\mathbf{p}_j$ is the Lagrangian multipliers corresponding to the price of item j. Based on the Lagrangian $\mathscr{L}$, the potential function becomes

$$\phi \equiv \max_{\{\mathbf{x}^i\}} \mathscr{L}(\{\mathbf{x}^i\}, \mathbf{p}). \tag{18}$$

Theorem 4 (Convexity and gradient of ϕ)**.** *ϕ defined in Equation (18) is convex and $\nabla\phi = -\mathbf{z}$, where $\mathbf{z}$ is defined in Equation (13) (proof in [30]).*

Gradient Descent Convergence Analysis. Given the excess demand potential ϕ, BANTER exploits the gradient descent to converge to the competitive equilibrium.

Theorem 5. *Denote $\mathbf{p}^* \equiv \arg\min_{\mathbf{p}} \phi(\mathbf{p})$ as the minimizer of the convex potential ϕ, and the optimal $\phi^* \equiv \phi(\mathbf{p}^*)$. Assume $\nabla\phi$ is L- Lipschitz continuous and step-size $\epsilon \leq \frac{2}{L}$. For any starting point $\mathbf{p}^0$, $\lim_{t\to\infty} \phi(\mathbf{p}^t) = \phi^*$ (proof in [30]).*

Proposition 2. *The pruning schemes OPPS and HPSS aim to effectively approximate equilibrium breeding utility. The inequality: $\frac{\sum_{\eta_j \in \mathcal{L}} V(\eta_j)}{\sum_{\eta_j \in \mathcal{M}} V(\eta_j)} \leq \left(\frac{K_c}{M}\right)^{1-1/\alpha}$ holds, where $K_c = |\mathcal{L}|$ and $M = |\mathcal{M}|$ (proof in [30]).*

To enhance computational efficiency, OPPS and HPSS prioritize promising parent breeding combinations in a smaller set $\mathcal{L}$, effectively trimming less valuable options from the full candidate pool $\mathcal{M}$. Proposition 2 then quantifies the quality of this approximation. Equipped with this bounded and stepwise approximation for breeding choices, BANTER ultimately converges to a competitive equilibrium (Theorem 5) because its iterative updates consistently minimize the overall excess demand potential ϕ derived from these buyer decisions.

VI. Experiments

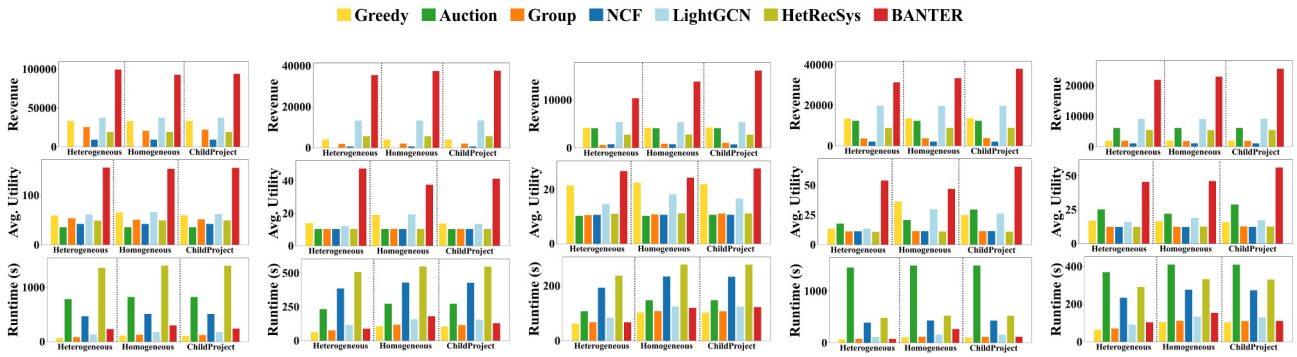
We experiment with five real-world NFT projects on the Ethereum blockchain: *i) Axie Infinity* [53] (5515 buyers and 26739 NFTs), *ii) Bored Ape Yacht Club (BAYC)* [54] (4483 buyers and 8141 NFTs), *iii) Crypto Kitties* [49] (1869 buyers and 5984 NFTs), *iv) Fat Ape Club* [23] (4540 buyers and 5189 NFTs), and *v) Roaring Leaders* [55] (2407 buyers and 4962 NFTs). These project, with trait systems and trade records up to November 17, 2023, are collected from OpenSea [11] using the Moralis Python SDK [56]. We leverage the trade records to assign buyers' budgets and preferences for each NFT project. In particular, we identify buyers by their blockchain addresses, filter out those with five or fewer NFT purchases, and exclude NFT instances without a trading history [19], [57]. The budget is calculated by summing the transaction prices of a buyer's purchases, while preferences are mined using [58] based on the traits in the acquired NFTs.

We compare BANTER against six baseline methods:[6] HetRecSys [19], [60], which models buyer-buyer, buyer-NFT, and NFT-NFT interactions; LightGCN [18], [61], which processes buyer-item interactions; NCF [62], which learns buyer and item embeddings; Group [63], [64], which recommends the same set of NFTs to groups of buyers (grouped by preferences); Auction [65], which leverages a bidding process; Greedy, which recommends NFTs with the highest value-to-price ratio (i.e., "*bang per buck*" [39]) for each buyer.

Each buyer is randomly set as a niche or eclectic collector for Heterogeneous Breeding. For BANTER, we set all iteration numbers to 128, (initial) step size $\epsilon = 1000, \epsilon_s = 1$, candidate length $K_C = 50$, mutation rate $r = 0.03$, breeding count $k = 10$. Due to the space constraint, qualitative tests to visualize the equilibrium is presented in [30].

[6]Following [59], the baselines set NFT prices proportional to their objective valuations. Please refer to [30] for detailed implementation.

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(a) *Axie Infinity* (b) *Bored Ape Yacht Club* (c) *Crypto Kitties* (d) *Fat Ape Club* (e) *Roaring Leader*

Fig. 4. Seller's revenue (top row), average buyers' utility (middle row), and runtime (bottom row) comparisons.

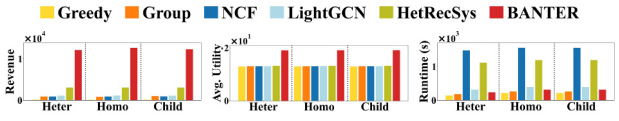


Fig. 5. Seller's revenue (left), average buyers' utility (middle), and runtime (right) on the Yelp dataset.

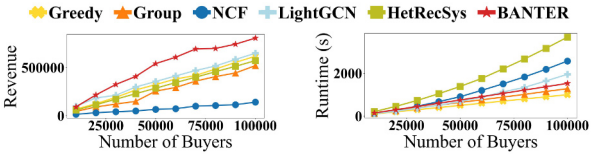


Fig. 6. Scalability test on large number of buyers.

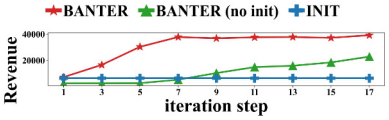


Fig. 7. Ablation tests over initial iteration steps.

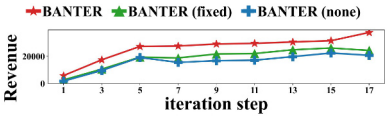


Fig. 8. Ablation tests on step-size scheduling.

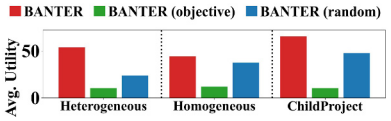


Fig. 9. Ablation tests on candidate sampling.

A. Experimental Results

We compare the performance of revenue (top), average buyer utility (middle), and runtime (bottom) for the three breeding mechanisms in Fig. 4. Across all metrics, BANTER consistently demonstrates superior performance, achieving the highest seller revenues and the highest average buyer utility, while incurring a low runtime. This performance improvement is attributed to BANTER's joint optimization of pricing for the seller and customized NFT purchasing and breeding recommendations for individual buyers. By closely approximating competitive equilibrium, BANTER optimizes these outcomes for all stakeholders, including the seller and all buyers.

Revenue: BANTER outperforms all baselines by a significant margin, iteratively optimizing pricing through PRICEREC to assign higher prices to in-demand NFTs. LightGCN often secures the second-best performance by modeling buyer preferences, enabling more precise identification of optimal purchases and reducing item collisions. However, it fails to account for breeding and cannot provide dynamic pricing based on buyer demands. HetRecSys generates less revenue compared to BANTER and LightGCN, as its overemphasis on buyer-buyer and NFT-NFT relations can obscure the direct individual demand signals vital for effective dynamic pricing. NCF performs the worst because it fails to account for the alignment between buyer preferences and NFT attributes. Auction exhibits unstable performance, as its bidding process forces buyers to exhaust their budgets to secure top choices, leading to erratic results. Group struggles by recommending the same NFTs to groups of buyers, increasing the likelihood of popular NFTs being sold out and missing revenue opportunities. Lastly, Greedy performs variably, depending on the concentration of buyer preferences, as it greedily exploits the amalgamated valuation to provide recommendations.

Utility: For buyer utility, BANTER consistently delivers superior results. Most baselines only slightly improve their initial utilities. This stagnation indicates baselines' failure to identify NFT purchasing opportunities where the utility gained from NFTs (comprised of $U_{Instance}$, $U_{Collection}$, and $U_{Breeding}$ (Definition 5)) exceeds the cost of the NFTs. In contrast, BANTER excels by carefully estimating the NFT breeding utilities while calculating buyer-specific $U_{Collection}$ and objective $U_{Instance}$, allowing for effective evaluation of purchasing each NFT instance. Furthermore, while BANTER dynamically adjusts NFT prices based on demand, it also optimizes every buyer's purchasing based on the updated price and additional breeding utilities, resulting in the most advantageous options for buyers, leading to substantial improvement in buyer utilities.

When comparing different breeding mechanisms, BANTER generally achieves lower utility with Homogeneous Breeding than with Child-project Breeding due to the population factor f_{pop}, which reflects market saturation. Besides, BANTER attains higher utility with Heterogeneous Breeding for NFT

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projects with a larger number of NFT instances (*Axie Infinity* and *Bored Ape Yacht Club*) since it provides more options for buyers to choose the best parent sets. Additionally, BANTER demonstrates greater performance gains in *Axie Infinity*, which has a larger number of buyers and NFTs, necessitating equilibrium between pricing and purchasing recommendations. In contrast, BANTER's advantage over baselines is less significant in *CryptoKitties*, which has fewer buyers and a simpler market structure.

Runtime: BANTER achieves a balanced runtime, outperforming computationally expensive methods such as HetRecSys, NCF, and Auction, which require extensive training, inference, or prolonged bidding processes. While LightGCN occasionally demonstrates lower runtimes, it falls short of delivering high-quality recommendations for both buyers and sellers. Group benefits from lower runtimes due to its group-based recommendation strategy, and Greedy achieves the lowest runtimes because of its simplistic design, which comes at the expense of suboptimal results. When comparing breeding mechanisms, BANTER shows a slightly higher runtime in Homogeneous Breeding due to the additional computational cost for deriving the population factor f_{pop}. While Heterogeneous Breeding involves combinations of more than three parent NFTs, HPSS effectively accelerates the computation of the breeding utility, allowing BANTER to maintain a low runtime. Consistent with the derivation of Theorem 2, runtimes scale linearly with the size of the NFT project.

B. Scalability Tests

We conduct scalability tests on 1) a real-world Yelp review dataset with 16045 users and 38031 items [66] and 2) a large-scale synthetic dataset prepared by duplicating the number of buyers ranging from $10,000$ to $100,000$ for *Fat Ape Club*.

First, to investigate the broader applicability of our proposed BANTER, we adopt the Yelp dataset [66] for the NP^{3}R problem to simulate an NFT project (see [30] for more details). The dataset comprises 16045 users as buyers and 38031 businesses as NFT instances. As shown in Fig. 5, BANTER consistently outperforms all baselines in all cases, demonstrating its applicability to larger NFT projects with rich attributes. This superior performance is attributed to BANTER's comprehensive modeling of a multifaceted buyer utility (capturing instance, collection, and breeding values), whose alignment with individual optima is grounded in the duality properties of homogeneous functions (Theorem 3).

Second, we evaluate the scalability of BANTER to a larger number of buyers. Fig. 6 presents a scalability test using synthetic data with the number of buyers N ranging from $10,000$ to $100,000$ and detailed in [30]. Specifically, the number of NFT instances fixed at $M = 5,000$, the preferences and budget of buyers are randomly set and the breeding mechanism is set to Child-project Breeding. As shown in Fig. 6, BANTER consistently outperforms all baseline methods in terms of the seller's revenue while maintaining a low runtime in all large-scale settings, linearly increasing with the number of buyers.

C. Ablation Tests

We conduct ablation tests on *Fat Ape Club* to reveal the importance of different designs in BANTER. First, we contrast the equilibrium process in BANTER against the preference-aware pricing initialization INIT (see Algorithm 1). Fig. 7 compares BANTER under Homogeneous Breeding with two ablation variants after running for a fixed number of initial iteration steps: i) BANTER$_{\text{no init}}$, which eliminates the INIT initialization, and ii) INIT, which directly uses the pricing obtained by INIT without jointly refining the pricing and purchasing recommendation towards equilibrium. Comparing BANTER with BANTER$_{\text{no init}}$, the results show that while INIT helps BANTER by accelerating the optimization process, the equilibrium process contributes more significantly to BANTER's success in attaining optimal results (Proposition 1).

Second, we investigate the effectiveness of *demand-aware scheduling* for the pricing recommendation by comparing the attained revenue after running for a fixed number of initial iteration steps under Homogeneous Breeding. Fig. 8 compares BANTER with: i) BANTER$_{\text{fixed}}$, which adopts a fixed decay rate for the step size, and ii) BANTER$_{\text{none}}$, which does not adjust initial step size. The results demonstrate that the demand-aware scheduling in PRICE-REC assists BANTER in achieving the highest revenue, as it effectively adjusts the step sizes based on excess demand to accelerate convergence. Note that similar trends are also observed under Child-project Breeding and Heterogeneous Breeding.

Besides, we evaluate the impact of candidate selection $\mathcal{L}$ obtained by OPPS and HPSS in NFT-REC for the purchasing recommendation under all three breeding mechanisms. Fig. 9 compares BANTER with two variants: i) BANTER$_{\text{objective}}$, which selects the NFT η with the highest $V(\eta)$, and ii) BANTER$_{\text{random}}$, which selects random candidates. As shown, personalization achieved by the amalgamated valuation $\tilde{V}$ significantly outperforms other approaches in terms of average buyer utility across all breedings, while the contrast between BANTER and BANTER$_{\text{random}}$ clearly indicates the efficacy of both OPPS and HPSS.

VII. Conclusion

To our knowledge, this paper first studies the NFT Project Pricing/Purchasing Recommendation (NP^{3}R) problem, highlighting challenges from its complex trait systems and diverse breeding mechanisms (Heterogeneous, Homogeneous, and Child-project). We propose BANTER (Breeding-aware NFT Equilibrium Recommendation) to address NP^{3}R, featuring iterative pricing (PRICE-REC) and purchasing/breeding (NFT-REC) components, with NFT-REC's efficiency enhanced by OPPS and HPSS pruning schemes. Theoretical analyses confirm NP^{3}R's PPAD-complete hardness and BANTER's convergence to competitive equilibrium; experiments on five real-world NFT datasets demonstrate BANTER significantly improves seller revenue and buyer utility with low runtime compared to baselines. Future work will explore time and social influences on NFT price fluctuations.

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